Review: Net-Ev^2: A Generative Simulator for Network Event Evolution

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1 Abstract

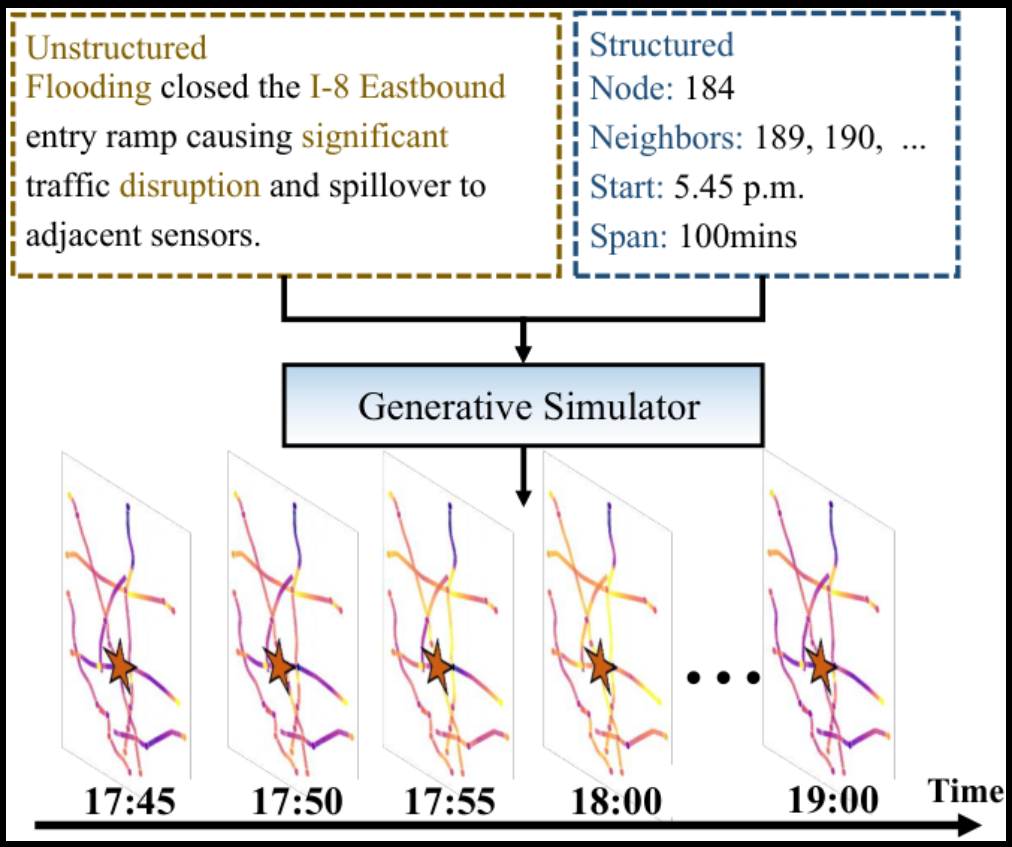
Reducing real-world trial and error has long been a central goal of decision making, and generative simulators advance this goal by modeling the evolution of future states. An even more challenging yet meaningful task is simulating how disturbance events (e.g., accidents) propagate their impacts across real-world networks. The existing approaches fall short of modeling both structured attributes and unstructured semantics of events, and capturing topological structures in simulating network event evolution. Therefore, we are motivated to propose Net-Ev2 (Network Event Evolution), a novel generative simulator that jointly leverages event cues while preserving network topology in simulations. Specifically, the framework consists of two stages, namely structure-guided masked pretraining and topology-aware diffusion process, which is achieved by U-Net-like graph downsampling and upsampling during denoising. At inference time, Net-Ev2can generate simulations using natural-language event input only, with greater flexibility for practical usage. Furthermore, we introduce Net-Ev2-6.5M, a multimodal benchmark of aligned event and network traffic data across four large-scale road networks, as well as a new topology-aware metric, namely JL-MMD, to evaluate topological fidelity in generated network dynamics. Extensive experiments demonstrate the state-of-the-art performance and strong generalization ability of Net-Ev2. Code is made available at https://github.com/Guangyu4/Net-Ev-2

2 Keywords

Generative Simulator, Network Event Evolution, Diffusion Models, Johnson-Lindenstrauss Lemma

3 Introduction

- Generative simulators → state evolution over time → closed-loop evaluation → scalable data generation
- Generative models come in two design paradigms:



1. Network to visual-like representations for diffusion models: line graphs, field transforms, or data rendered as RGB images. Struggles to capture full complexity of network evolution
2. Diffusion architechtures: flow matching, denoising diffusion probabilistic models. Better fit for variable length sequences, multivariate dependencies, long-term trends etc.

- Network evolution poses challenges for existing methods, and events can have textual and quantitative modalities
- **Example:** unstructured text (I-8 Eastbound and Significant Traffic), quantitiative attributes such as node indices and timespans
- Problematic to combine modalities, language encoders such as BERT struggle with numerical data as input
- Structured data harder to obtain than unstructured data in real-wordld applications, and ideally model should use structure information for training but perform well in the absence of it
- Networks have graph topology, events propogate on edges. Denoiser models fail to capture propogation details
- This issue rarely considered in evaluation, topology-aware metrics are largely missing

- Paper proposes Net-Ev^2 generative simulator for network event evolution and associated multimodel benchmark (6.5M events)
- introduce topology aware metric JL-MMD, inspired by Johnson-Lindenstrauss lemma

Contributions:

- Problem: simulating evolution of network states under textual event conditions
- Propose the Net-Ev^2 generative model, which combines masked pretraiing with a topology-aware denoiser
- Net-Ev^2-6.5M a multi-modal benchmark of aligned event text and network traffic data
- JL-MMD a metric based on Johnson-Lindenstrauss random projection to evaluate topological fidelity on generated network dynamics

4 Preliminaries

Model a network graph as $G = (V, A)$ where V is a set of N sensor nodes and $A \in \mathbb{R}^{N \times N}$ is the adjacency matrix representing the network topology. $X \in \mathbf{R}^{T \times N}$ denote a sequence of network snapshots (e.g. traffic volume) over T steps. For each efvent X is represented by a set of multi-structural attributes $E = E_{i_{i=1}}^{M}$, including a textual description E_d and structured properties. Given these conditions as an initial state, the problem is to simulate realistic future states $\hat{X}$ that reflects how an even evolves and propogates over the network topology.

$$E = E_{i_{i=1}}^{M} \overset{G}{\rightarrow} \hat{X} \in \mathbf{R}^{T \times N}$$

E is comprised of $E = (E_d, E_n, E_t)$, corresponding to the textual description, spatial context, and impacted nodes in the egocentric subgraph (whatever that means), and temporal context, respectively.

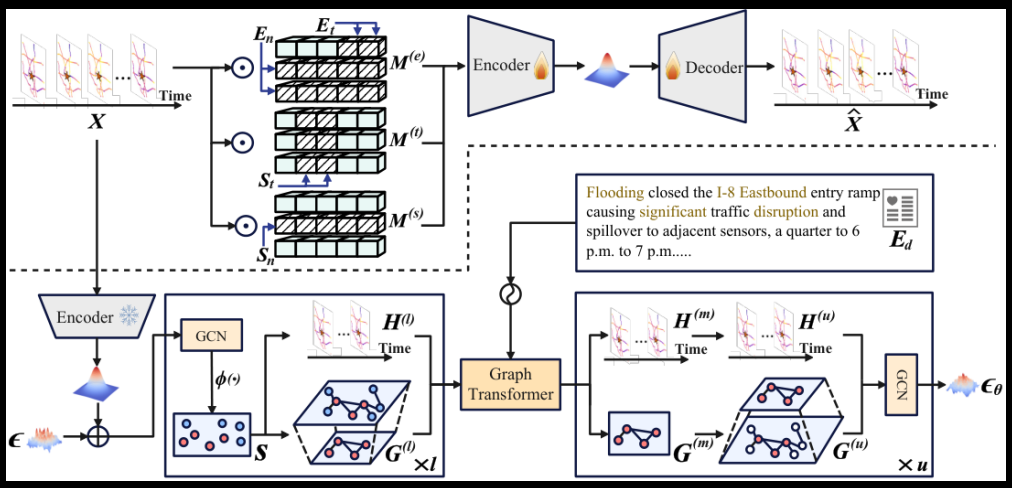
5 Net-Ev2-6.5M Benchmark

New multimodal benchmark integrating signal observations from LargeST [4] and event records from LSTW [5]. Covers San Diegeo, Greated Bay Area, Greater Los Angeles, and California. Previous little work combining traffic and event datasets, so matched using the following conditions

1. *Event-traffic alignment* Align sensor readings to incident reports and weather events
2. *Event descriptions* Attach weather descriptions to weather events using LLMs matching style as LSTW incident events
3. *Multi-structural sample organization* 6.5M samples consisting of (1) textual description E_d, event descrioptoin encoded by text encoder; (2) network snapshots $X \in \mathbf{R}^{T \times N}$, the subsequent time steps, (3) Spatial context E_n, the target node and its n-hop neighbors over the network; Teemporat context E_t the event's start and end timestamp

6 Methodology

Input is a network snapshot sequence X, fed into encoder with spatio-temporal and structure guided masks, following a masked autoencoder design. Noise added to hidden representations during training, and topology aware denoiser removes it using pooling/unpooling operations to preserve network topology. Unstructured even conditions are injected vis AdaLN [1]



6.1 Structure guided MAE

MAE compresses traffic observations to latent space where diffusion operates efficiently. During training use multi-structural event information to guide masking strategies. Apparently a better method than encoding as special tokens.

6.1.1 Spatio-temporal masking

Binary masks $\mathbf{M}^{(t)}, \mathbf{M}^{(s)} \in \{-0, 1\}^{T \times N}$ by selecting random subsets of $\mathcal{T}\lfloor \rho \cdot T$ and $\mathcal{S}\lfloor \rho \cdot S$ and setting masks to zero for both.

6.1.2 Structure-guided masking

Masking nodes E_n and corresponding time windows E_t:

$$\mathbf{M}^{(e)}_{:,j} = 0 \text{ for } j \in E_n \qquad \mathbf{M}^{(e)}_{i,:} = 0 \text{ for } i < \max(E_t)$$

where E_t denotes the events of temporal span, and i $< \max(E_t)$ masks the entire event duration

6.1.3 Variational encoder-decoder architechture

adopts a variational formulation [3] for generalizability. Encode masked input to latent representation where decoder reconstructs from original signal. The encoder projects the input to latent space then applies transformer layers:

$$[\mu, \log\sigma^2] = \text{TransformerEncoder}(\text{Linear}(X \odot M))$$

Where we then sample $z = \mu + \sigma \odot \xi$ with $\xi \sim \mathcal{N}(0, \mathbf{I})$ during training, and use $z = \mu$ at inference. The decoder then maps back z to original space $\hat{X} \in \mathbf{R}^{T \times N}$

6.1.4 Training Objective

Variational encoder-decoder is fed temporal, spatial, and structure-guided masked inputs and outputs a final result fused by residual addition $\hat{X} = \hat{X}^{(t)} + \hat{X}^{(s)} + \hat{X}^{(e)}$. Model is optimzed on a loss function that combines reconstruction accuracy and KL-divergence regularization:

$$\mathcal{L}_{\text{MAE}} = \|\mathbf{X} - \hat{\mathbf{X}}\|_2^2 a + w \sum_{r \in \{t,s,e\}} D_{\text{KL}}(q_r(\mathbf{z}|\mathbf{X}, \mathbf{M}^{(r)}) \| (\mathbf{z}))$$

where $p(\mathbf{z} = \mathcal{N}(0, \mathbf{I}))$ is the prior distribution and each D_{KL} term regularizes the latext space to smooth for subsequent diffusion.

6.2 Topology-aware diffusion moddel

6.2.1 Diffusion process

Adopts DDPM [2] method in latent space. For latent representation $\mathbf{z}_0$ forward process adds Gaussian noise $q(\mathbf{z}_t|\mathbf{z}_0)$ with noise schedule $\{\beta_t\}$. The reverse process learns to denoise conditioned on $\mathbf{c}$:

$$p\theta(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{c}) = \mathcal{N}(\mathbf{z}_{t-1}, \mu_\theta(\mathbf{z}_t, t, \mathbf{c}), \widetilde{\beta}_t, \mathbf{I})$$

6.2.2 Graph U-Net architecture

Event conditions are incorporated into Graph U-Net to preserve network topology, since eah Graph U-Net block propagates information and encourages model to learn inter-node relationshiops. Graph U-Net takes $(\mathbf{z}_t, t, \mathbf{c})$ inputs and predicts the noise ϵ_θ. Consists of encoder-bottleneck-decoder with L levels and skip connections. Encoder pools graph from $G^{(0)}$ to $G^{(L)}$, bottleneck injects conditioning, and decoder pools back to original topology.

Graph Pooling Learnable pooling used to build hierarchical representations while preserving topology. Projection later computes node importance score $\pi = \text{sigmoid}(\text{Linear}(\mathbf{H}^{(l)})) \in \mathbf{R}^{N_l}$, and top-$k$ nodes selected $\mathcal{J} = \text{top-}k(\pi)$. Pooled features are then constructed as $\mathbf{H}^{l+1} = \mathbf{H}^{(l)}\mathcal{J} \odot \pi\mathcal{J}$, where $\pi\mathcal{J}$ serves as a gating mechanism, emphasizing informative nodes.

Graph Transformer Block At bottleneck, injects conditioning through AdaLN [1]. Different from standard layer normalization, AdaLN derives scale, shift, and gate, γ, δ, λ respectively, from conditioning vector $\mathbf{c}$:

$$[\gamma, \delta\, \lambda] = \text{Linear}(\mathbf{c}), \quad \textbf{AdaLN}(\mathbf{H}, \mathbf{c}) = \gamma \odot \text{LayerNorm}(\mathbf{H}) + \delta$$

Each sub-layer applies AdaLN before its operation and uses the gate λ so scale the residual, GCN$(\cdot)$ denotes a graph convolutional network, and Attention corresponding to standard transformer self-attention module:

$$\begin{aligned}\mathbf{H}_g &= \mathbf{H}^{(L)} + \lambda_g \odot \text{GCN}(\text{AdaLN}(\mathbf{H}^{(L),\mathbf{c}}), \mathbf{A}^{(L)}) \\ \mathbf{H}_a &= \mathbf{H}_g + \lambda_a \odot \text{Attention}(\text{AdaLN}(\mathbf{H}_g, \mathbf{c})) \\ \mathbf{H}_m &= \mathbf{H}_a + \lambda_m \odot \text{MLP}()\text{AdaLN}(\mathbf{H}_a, \mathbf{c})\end{aligned}$$

Graph Unpooling decoder restores graph resolution by placing pooled features back to original positions using $(\mathbf{H}^{(u-1)}, G^{(u-1)}) = \text{unpool}(\mathbf{H}^{(u)}, \pi^{(u)}, \mathbf{A}^{u-1})$, where GCN layers propogating information to unselected nodes.

6.2.3 Conditionaing Mechanism

Condition diffusion on timestep and event text embedding $\mathbf{E}_d$. Existing practice [1, 7]. Time steps encoded using sinusoidal positional embeddings, text embedding obtained vie pre-trained BERT and combined to create conditioning vector $\mathbf{c}$. Structured attributes guide representation learning from the masking objective but not required by denoiser at inference, allowing for scenario simulate from natural-langauge alone.

6.2.4 Training Objective

The denoiser is trained to predict noise added at each timestep:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t,\mathbf{z}_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(\mathbf{z}, t, c)\|_2^2\right]$$

6.2.5 Infrence

Start with Gaussian noise $\mathbf{z}_{T_{\text{diff}}} \sim \mathcal{N}(0, \mathbf{I})$ and iteratively denoise:

$$\mathbf{z}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\Big(\mathbf{z}_t - \frac{1-\alpha_t}{\sqrt{q-\bar{\alpha}_t}}\epsilon_\theta(\mathbf{z}_t, t, \mathbf{c})\Big) + \sqrt{\widetilde{\beta}_t}\epsilon$$

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ for $t > 1$ and $\epsilon = 0$ for $t = 1$. The final $\mathbf{z}_0$ is decoded to obtain $\hat{\mathbf{X}}$.

7 Experiments

7.1 Experimental setup

Implementation Details

- PyTorch 2.3.1+cu121
- Python 3.10
- Four NVIDIA RTX A800 GPUs
- Adam optimizer initial learning rate of 0.0001
- Samples split chronologically, 80%/20% Train/Test Split

Baselines

- Diffusion-TS
- VerbalITS
- ChatTraffic
- T2S

Evaluation Metrics

- Euclidean distance: Point wise distance between generated and ground truth sequences
- Context Fréchet Inception Distance: Semantic distance of latent space
- Contrastive Time series Text Pretraining: CLIP-style cross-modal retrieval
- Johnson-Lindenstrauss Maximum Mean Discrepancy (JL-MMD)

8 Case Studies

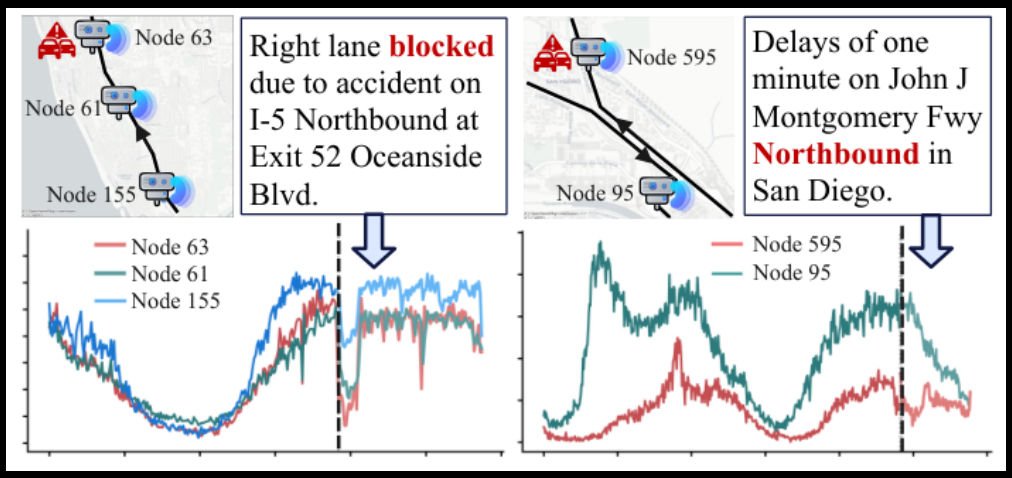
Case study (a) shows event propogation between subsequent nodes and direction-specific disruptions
Case study (b) shows varied intensities based on natural-language input

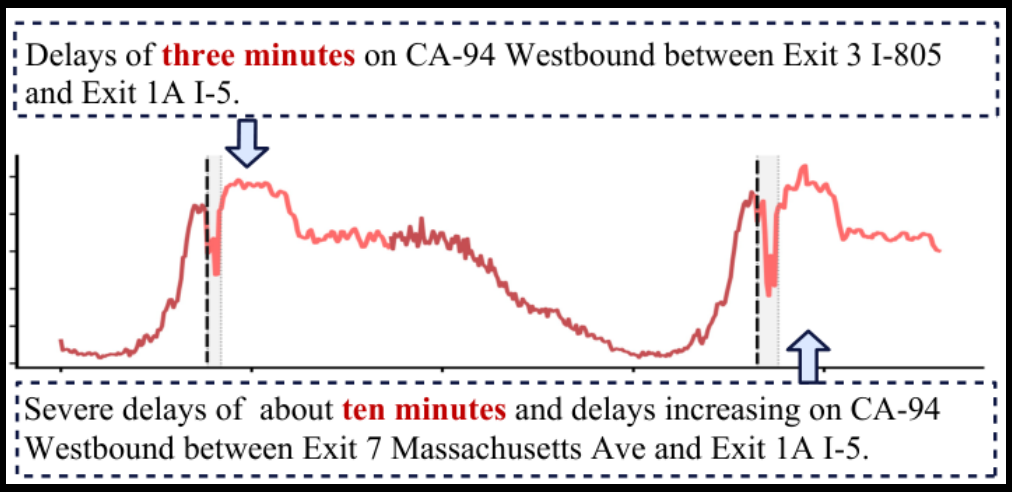
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